NAME: ______________________________ **PERIOD:** ________

Which Regression Line Fits?

AP Statistics · Unit 5 · Topic 5.5

[STAR] THE PROBLEM

Suppose a solar-farm analyst uses sunshine hours x to predict daily generation y in megawatt-hours (MWh). The table summarizes 12 observed days.

Quinn: "Using $b = r(s_y/s_x)$ gives $\hat{y} = 5.6 + 3.2x$. It passes through $(7, 28)$, so it must be the LSRL. It explains 80 percent of the variation."

Rowan: "Using $b = r(s_x/s_y)$ gives $\hat{y} = 26.6 + 0.2x$. It also passes through $(7, 28)$, so it is equally good. Explained variation is 64 percent."

Solar-farm summary (12 observed days)

x range	$\bar{x}$	$\bar{y}$	s_x	s_y	r
4–10 h	7 h	28 MWh	1.5 h	6 MWh	0.80

FIRST TAKE — one sentence before you work the parts

The two lines agree at 7 sunshine hours. Does that make them equally useful at other times? Give your first thought in one sentence.

[BOOK] MAKE THE REGRESSION PROPERTIES AGREE (keep on the page)

The least-squares regression line (LSRL) minimizes the sum of squared residuals and contains $(\bar{x}, \bar{y})$. Its slope is $b = r(s_y/s_x)$, its intercept is $a = \bar{y} - b\bar{x}$, and r^2 is the proportion of response variation explained by the linear model. Interpret slope as a predicted response change per unit of x. The intercept predicts at $x = 0$, which may be outside the observed range and unsupported in context.

Work (a)–(c).

DOK 1 **(a)** State the mean point that every least-squares regression line must contain.

DOK 2 **(b)** Calculate the regression slope and intercept, and r^2. Give the slope's units.

DOK 3 **(c)** Which parts of Quinn's and Rowan's claims should a report keep? Use the slope formula to choose a line, explain why their shared mean point does not make both lines correct, and interpret the correct percent of variation explained. Write 3–4 sentences.

> **Frame:** I would keep ________________ from Quinn and ________________ from Rowan because ________________.

> **Frame:** Sharing $(7, 28)$ is not enough because ________________; the model explains ________ percent of variation in ________________.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.